The core database for prokaryotic strain data

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German Collection of Microorganisms and Cell Cultures

GLOBAL CORE BIODATA RESOURCE

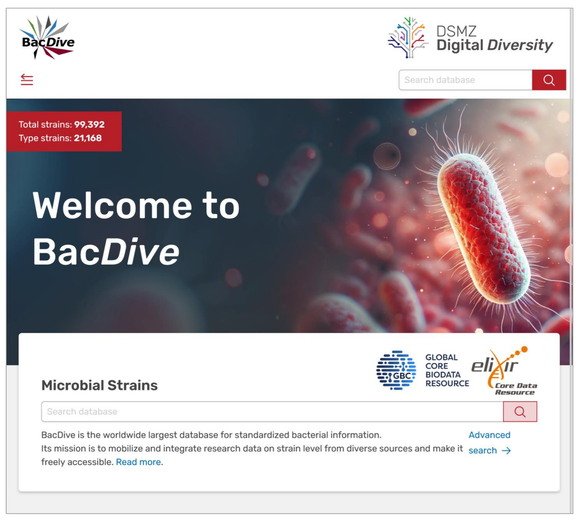
elixir Core Data Resource

THE BACTERIAL DIVERSITY DATABASE

BAC*DIVE* AND ITS MISSION

Bac*Dive* is the **worldwide largest database** for standardized bacterial information.

Its mission is to gather and standardize phenotypic **strain-level research data** from diverse sources including internal catalogs of culture collections and primary literature to make them freely and easily accessible.



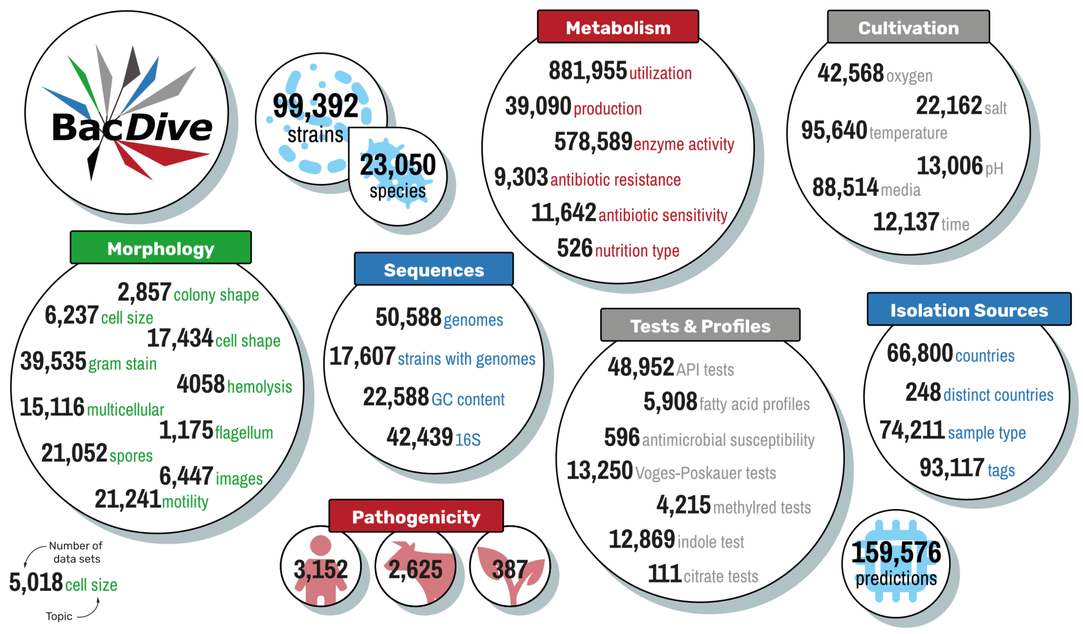
Read and cite the newest publication!

Schober I, Koblitz J, Carbasse JS, Ebeling C, Schmidt ML, Podstawka A, et al. **Bac*Dive* in 2025: the core database for prokaryotic strain data.** *Nucleic Acids Research* 2024; 53: D748–D756

RELEASE 2024

CONTENT IN NUMBERS

Total strains: **99,392**
Data points: **2,709,516**



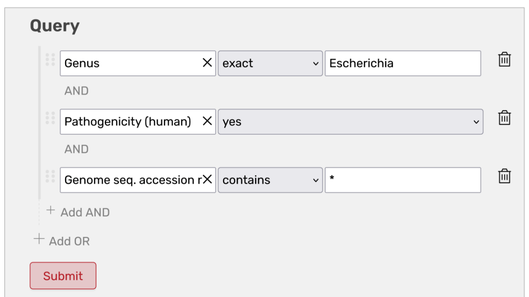
DATA QUERIES

SEARCHING CONTENT

Easily find strains of interest

Bac*Dive* tools are focused on **searching and retrieving** data.

The *Simple Search* was recently supplemented with a smart feature, now making **search suggestions** based on precalculated *Advanced Search* queries.

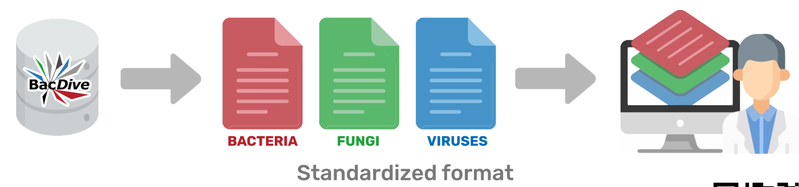


The *Advanced Search* allows for large-scale queries **combining several data fields** for comparative analyses of multiple strains.

STRAIN DATA STANDARD

STRUCTURING CONTENT

Bac*Dive* will provide a standard for the exchange of strain-related data. This will facilitate easier **sharing, comparing and integrating** of data.



An open structure for data exchange

https://leibnizdsmz.github.io/microbial-data-standard

GENOME-BASED PREDICTIONS

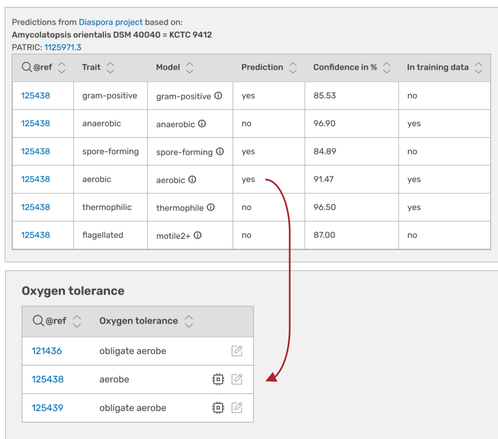
FILLING CONTENT GAPS

DiASPora diaspora-project.de

Large gaps remain in the phenotypic description of many strains. In order to decrease some of them, genome-based predictions were recently introduced into Bac*Dive*.

All predicted data are listed in a new section on the strain detail page with confidence values.

Predicted data points with over 90% confidence are additionally integrated into the other sections marked with their confidence value and an AI icon.



Strain-level phenotype predictions were produced using machine-learning models trained on public genome sequences and high-quality standardized data from Bac*Dive*.

95,628 data points were generated by genome-based predictions **covering 15,938 strains**. High-confidence predictions have added completely **new information for 15,977 data points** for which no experimental data is available.

Koblitz J, Reimer LC, Pukall R, Overmann J. **Predicting bacterial phenotypic traits through improved machine learning using high-quality, curated datasets.** *bioRxiv* 2024.

THE BAC*DIVE* KNOWLEDGE GRAPH

SEMANTIC INTEGRATION OF CONTENT

The new knowledge graph of the Bac*Dive* database contains **more than 16.5 million triples** and provides a powerful SPARQL endpoint to directly search and analyze the knowledge provided through Bac*Dive* in a standardized way. This is supported by a **new descriptive ontology**, which provides a structured framework to represent the various aspects of microbial data.

More on **Poster 74**

NEW BAC*DIVE* DESIGN

PRESENTING CONTENT

Try it out and leave feedback!

A new design for the Bac*Dive* website is currently in beta testing phase.

The new look follows the DSMZ Digital Diversity design with the aim of providing a **consistent user experience** across all DSMZ databases. The wealth of information BacDive contains for each strain is now also **better structured** in different thematic tabs.

beta.bacdive.dsmz.de

DSMZ Digital Diversity

PARTNERS

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